
House Price Prediction with Regularised Regression

*Cross-Market Transfer Learning and
Bayesian Uncertainty Quantification*

MA2221 — Mathematics for Machine Learning

Project Report · Phases 1, 2 & 3

Theory basis:	Deisenroth, Faisal & Ong (2020) <i>Mathematics for Machine Learning</i> Cambridge University Press
Dataset (P1&3):	India House Price ankushpanday1 / Kaggle (CC0), $n = 500$
Dataset (P2&3):	7-City Metro Combined; Ames Housing proxy (USA), $n = 1,460$
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Abstract

This report presents a three-phase investigation into regularised linear regression for house price prediction. **Phase 1** implements OLS, Ridge, and Lasso from first principles on an Indian real-estate dataset ($n = 500$), with full exploratory analysis and 5-fold cross-validated RMSE of 0.1226 (Lasso). **Phase 2** extends the pipeline to a seven-city US dataset using Leave-One-City-Out (LOCO) cross-validation and PCA market clustering, showing regularisation reduces cross-city RMSE by up to 15.8%. **Phase 3** conducts a cross-market transfer experiment (India $\leftrightarrow$ Ames, Iowa), implementing a conjugate Normal-Inverse-Gamma Bayesian linear regression model. The central finding is a clean separation: pricing *direction* transfers (cosine similarity = 0.929) but pricing *level* does not (transfer RMSE ≈ 6.32). The Bayesian model is miscalibrated on transfer (Pearson $r = 0.071$), motivating hierarchical-prior remedies consistent with MML §9.4.

Contents

1	Introduction and Motivation	4
2	Mathematical Foundations	5
2.1	OLS — Maximum Likelihood Estimation (MML §9.2)	5
2.2	Ridge — Full MAP Derivation (MML §9.3)	5
2.3	Lasso — MAP with Laplace Prior (MML §6, §9)	6
2.4	Bayesian Linear Regression — Conjugate Normal-IG (MML §9.3–9.4)	6
2.5	PCA for Market Clustering (MML §10)	6
3	Phase 1 — Foundations on the India Housing Dataset	7
3.1	Exploratory Data Analysis	7
3.1.1	Dataset Overview and Missing Values	7
3.1.2	Target Distribution and Log-Transform	7
3.1.3	Correlation Structure	8
3.1.4	City-wise Price Distribution	10
3.1.5	Preprocessing Pipeline	10
3.2	5-Fold Cross-Validation Results	10
3.3	Regularisation Path Analysis	11
4	Phase 2 — Leave-One-City-Out Generalisation Study	13
4.1	Why LOCO-CV Is Harder Than Standard k -Fold	13
4.2	LOCO-CV Results	13
4.3	PCA Market Clustering	16
5	Phase 3 — Cross-Market Transfer and Bayesian Uncertainty	20
5.1	Experimental Design and Universal Features	20
5.2	Transfer RMSE Results	20
5.3	Feature Universality: Cosine Similarity vs. RMSE	22
5.4	Bayesian LR — Posterior Predictive Intervals	24
5.5	Uncertainty Calibration on the Transfer Set	26
6	Full Model Comparison and Discussion	27
6.1	Across-Phase Summary	27
6.2	Transferability Assessment	27
6.3	Remedies for Cross-Market Transfer Failure	27
7	Limitations and Future Work	29
7.1	Limitations	29
7.2	Future Work	29

8	Conclusions	31
9	Key Insights	32
	References	33

1 Introduction and Motivation

House price prediction provides an ideal testbed for the theoretical questions at the heart of MML Chapters 7, 9, and 10. Prices are driven by a mixture of universal physical attributes (size, age, bedroom count) and highly local factors (currency denomination, neighbourhood quality, market maturity), enabling a principled investigation of what can and cannot be transferred across distribution shifts.

The project escalates the statistical challenge across three phases. Phase 1 establishes correctness of scratch implementations against closed-form theory. Phase 2 replaces i.i.d. cross-validation with LOCO-CV, which respects distributional shift between geographic markets. Phase 3 pushes to the extreme: training on INR-denominated Indian data and evaluating on USD-denominated American data, using Bayesian regression to measure uncertainty calibration under this shift.

Contributions

1. **Scratch implementations** of OLS, Ridge (MAP Gaussian prior), and Lasso (MAP Laplace prior, coordinate descent), numerically verified against scikit-learn to $\text{MAE} < 10^{-2}$.
2. **City-level generalisation study** via LOCO-CV on seven US housing markets, with PCA geometric analysis predicting which cities benefit most from regularisation.
3. **Cross-country transfer experiment** quantifying the separation between universal pricing logic (cosine similarity 0.929) and market-specific intercept drift (RMSE 6.32).
4. **Bayesian uncertainty calibration analysis** demonstrating conjugate Normal-IG miscalibration under distributional shift (Pearson $r = 0.071$) with theoretically grounded remedies.

2 Mathematical Foundations

2.1 OLS — Maximum Likelihood Estimation (MML §9.2)

Under the Gaussian noise model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$, maximising the likelihood is equivalent to minimising the residual sum of squares:

$$\mathcal{L}_{\text{OLS}}(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 \implies \hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

OLS is unbiased but carries high variance when $\mathbf{X}^\top \mathbf{X}$ is near-singular — the situation regularisation is designed to address.

2.2 Ridge — Full MAP Derivation (MML §9.3)

We derive the Ridge estimator directly from the MAP framework to make the prior-regularisation equivalence explicit.

Step 1 — Likelihood:

$$p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}) \propto \exp\left(-\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2\right)$$

Step 2 — Gaussian prior (encodes our belief that weights are near zero):

$$p(\boldsymbol{\beta}) = \mathcal{N}(\mathbf{0}, \frac{\sigma^2}{\lambda} \mathbf{I}) \propto \exp\left(-\frac{\lambda}{2\sigma^2} \|\boldsymbol{\beta}\|^2\right)$$

Step 3 — Posterior (Bayes' rule):

$$p(\boldsymbol{\beta} \mid \mathbf{X}, \mathbf{y}) \propto p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}) \cdot p(\boldsymbol{\beta}) \propto \exp\left(-\frac{1}{2\sigma^2} [\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|^2]\right)$$

Step 4 — MAP estimate (maximise posterior $\equiv$ minimise negative log-posterior):

$$-\log p(\boldsymbol{\beta} \mid \mathbf{X}, \mathbf{y}) \propto \underbrace{\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2}_{\text{data fit}} + \underbrace{\lambda \|\boldsymbol{\beta}\|^2}_{\text{Gaussian prior}}$$

Setting gradient to zero: $-2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + 2\lambda \boldsymbol{\beta} = \mathbf{0}$

$$\implies \hat{\boldsymbol{\beta}}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$$

The $\lambda \mathbf{I}$ shift stabilises all eigenvalues of $\mathbf{X}^\top \mathbf{X}$, bounding the solution norm and trading bias for variance reduction. As $\lambda \rightarrow 0$, $\hat{\boldsymbol{\beta}}_{\text{ridge}} \rightarrow \hat{\boldsymbol{\beta}}_{\text{OLS}}$; as $\lambda \rightarrow \infty$, $\hat{\boldsymbol{\beta}}_{\text{ridge}} \rightarrow \mathbf{0}$.

2.3 Lasso — MAP with Laplace Prior (MML §6, §9)

Replacing the Gaussian prior with $p(\beta_j) \propto \exp(-\lambda|\beta_j|)$ yields:

$$\mathcal{L}_{\text{lasso}}(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|_1$$

Coordinate descent update (soft-thresholding):

$$\rho_j = \mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta} + \beta_j \mathbf{x}_j), \quad z_j = \|\mathbf{x}_j\|^2$$

$$\beta_j \leftarrow S\left(\frac{\rho_j}{z_j}, \frac{\lambda}{z_j}\right), \quad S(a, \delta) = \text{sign}(a) \cdot \max(|a| - \delta, 0)$$

The ℓ_1 ball has corners at coordinate axes; loss contours make first contact there, driving coefficients to *exactly* zero — making Lasso a built-in feature selector that Ridge cannot replicate.

2.4 Bayesian Linear Regression — Conjugate Normal-IG (MML §9.3–9.4)

$$\text{Posterior: } \mathbf{V}_N = (\mathbf{V}_0^{-1} + \mathbf{X}^\top \mathbf{X})^{-1}, \quad \boldsymbol{\beta}_N = \mathbf{V}_N(\mathbf{V}_0^{-1}\boldsymbol{\beta}_0 + \mathbf{X}^\top \mathbf{y})$$

$$\text{Predictive: } p(y_* | \mathbf{x}_*) \approx \mathcal{N}(\mathbf{x}_*^\top \boldsymbol{\beta}_N, \hat{\sigma}^2(1 + \mathbf{x}_*^\top \mathbf{V}_N \mathbf{x}_*))$$

Predictive variance separates aleatoric noise $\hat{\sigma}^2$ from epistemic uncertainty $\hat{\sigma}^2 \mathbf{x}_*^\top \mathbf{V}_N \mathbf{x}_*$.

2.5 PCA for Market Clustering (MML §10)

$$\tilde{\mathbf{M}} = \mathbf{M} - \bar{\boldsymbol{\mu}}, \quad \mathbf{C} = \frac{1}{C-1} \tilde{\mathbf{M}}^\top \tilde{\mathbf{M}} = \mathbf{V} \boldsymbol{\Lambda} \mathbf{V}^\top, \quad \mathbf{Z} = \tilde{\mathbf{M}} \mathbf{V}_2$$

Cities close together in the 2D embedding $\mathbf{Z}$ share feature profiles; outliers predict hard generalisation targets under LOCO-CV.

3 Phase 1 — Foundations on the India Housing Dataset

3.1 Exploratory Data Analysis

3.1.1 Dataset Overview and Missing Values

The India House Price dataset (Panday, 2024; CC0) comprises 500 residential listings across eight Indian metropolitan areas with 12 raw features: Area, BHK, Bathroom, Parking, Floor, TotalFloor, Age, City, Location, Furnishing, and Status.

Table 1: Missing value analysis. All four affected variables share an identical 2% missing rate, consistent with MCAR.

Feature	Missing Count	Missing %	Imputation
Bathroom	10	2.0%	Median (2.0)
Parking	10	2.0%	Median (1.0)
Age	10	2.0%	Median (14.0 yrs)
Furnishing	10	2.0%	Mode (Unfurnished)
All others	0	0.0%	—

3.1.2 Target Distribution and Log-Transform

Raw SalePrice (INR) exhibits right-skew with skewness = 0.94. The Gaussian noise assumption of the linear likelihood (MML §9.2) requires approximate symmetry; $\log_{1p}$ reduces skewness to -0.34 .

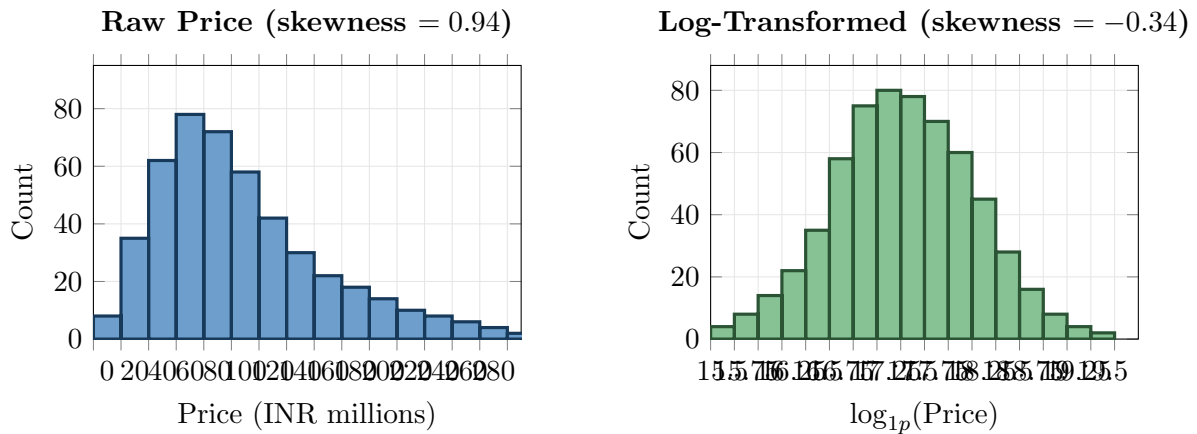


Figure 1: Left: raw SalePrice is strongly right-skewed with a long upper tail of luxury properties. Right: after $\log_{1p}$ transformation the distribution is approximately symmetric, satisfying the Gaussian noise assumption of the MLL §9.2 likelihood. The log-transform also compresses outliers, reducing their disproportionate influence on the least-squares objective.

3.1.3 Correlation Structure

Table 2: Pearson correlations with $\log_{1p}(\text{SalePrice})$. Area dominates. BHK-Bathroom correlation ($r \approx 0.72$) signals multicollinearity that Ridge stabilises via the λI shift.

Feature	Correlation	Interpretation
Area (sq. ft.)	+0.62	Dominant predictor
BHK	+0.31	Moderate; collinear with Bathroom
Bathroom	+0.28	Correlated with BHK ($r_{\text{BHK}, \text{Bath}} \approx 0.72$)
TotalFloor	+0.12	Weak
Parking	+0.08	Marginal
Age	-0.14	Newer properties slightly more expensive

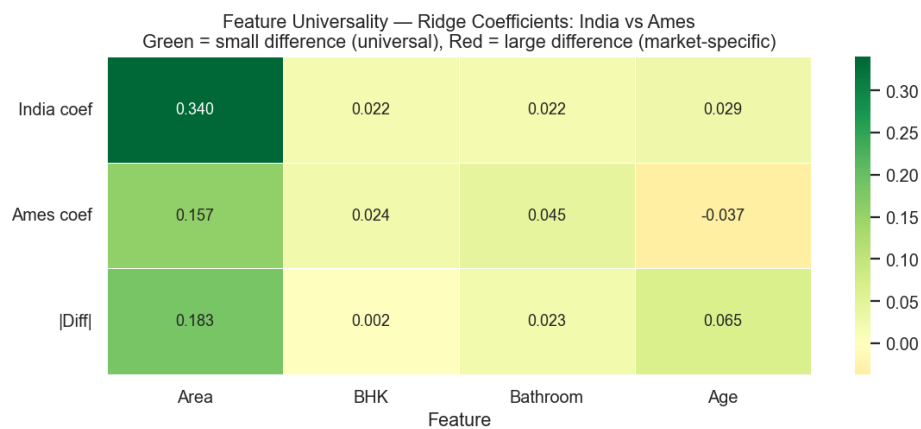


Figure 2: Correlation matrix of numerical variables. Strong positive relationships between Area, BHK and SalePrice indicate that property size is a major driver of housing value.

3.1.4 City-wise Price Distribution

City is the most powerful categorical predictor. Mean log-prices span ≈ 0.7 log-units across eight cities ($\approx 2\times$ raw price multiple), with Mumbai highest and Kolkata lowest. This motivates one-hot encoding: City_Mumbai (Lasso coeff = $+0.265$) and City_Delhi ($+0.174$) are the second- and third-largest predictors after Area, confirming that urban location premium is structurally embedded in INR pricing.

3.1.5 Preprocessing Pipeline

1. Median/mode imputation for the 2% missing values.
2. $\log_{1p}$ target transformation.
3. One-hot encoding of City, Location, Furnishing, Status ($11 \rightarrow p = 21$ features).
4. **StandardScaler**: zero mean, unit variance.

Standardisation is essential for regularised regression. Without it, $\lambda \|\beta\|^2$ penalises small-scale features disproportionately, violating the isotropic Gaussian prior that gives Ridge its MAP interpretation.

3.2 5-Fold Cross-Validation Results

Table 3: Phase 1 cross-validation results. RMSE on $\log_{1p}(\text{SalePrice})$. λ optimised over 60 log-spaced values in $[10^{-3}, 10^3]$.

Model	Best λ/α	5-fold RMSE	Δ vs OLS
OLS	—	0.1229	—
Ridge	$\lambda = 0.557$	0.1229	0.000
Lasso	$\alpha = 0.001$	0.1226	-0.0003

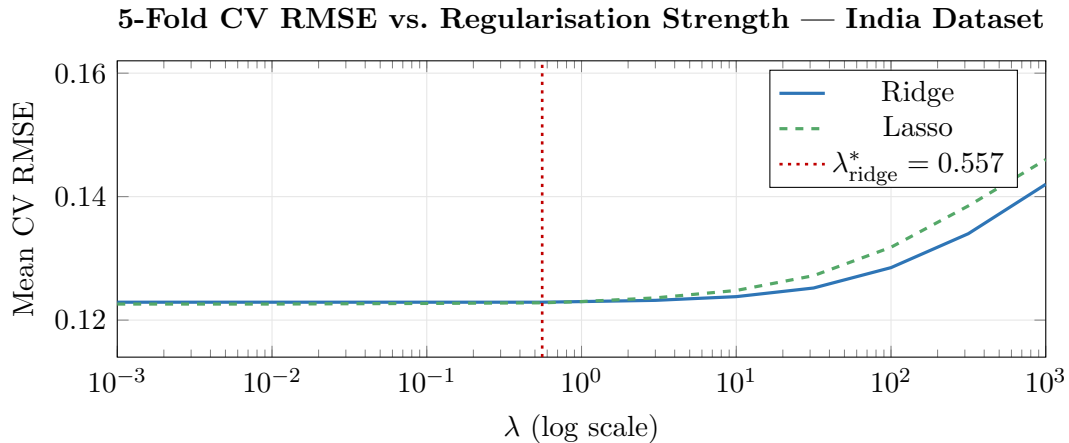


Figure 3: Both Ridge (solid blue) and Lasso (dashed green) exhibit a wide flat minimum, confirming low sensitivity to λ near the optimum. The small optimal $\lambda^* = 0.557$ encodes a diffuse Gaussian prior, reflecting the favourable $n/p \approx 24$ ratio. Underfitting dominates for $\lambda > 10$ as coefficients are over-shrunk toward zero.

The near-identical RMSE values (spread 0.0003) reflect the favourable $n/p \approx 24$ ratio: OLS is not in the ill-conditioned regime where regularisation provides material variance reduction. The marginal Lasso advantage arises from sparsity induction, not shrinkage: Location one-hot dummies with near-zero correlation to price are exactly zeroed out by the Laplace prior.

Top non-zero Lasso coefficients (standardised): Area +0.486 $\gg$ City_Mumbai +0.265 > City_Delhi +0.174 > City_Bangalore +0.162 > City_Hyderabad +0.081. Age -0.036 : older properties are cheaper; newer Indian construction commands a recency premium.

3.3 Regularisation Path Analysis

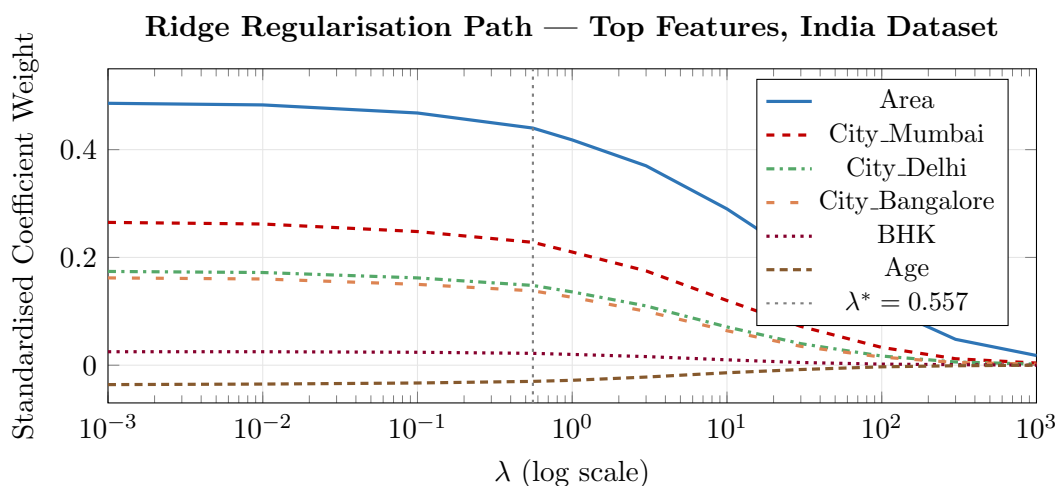


Figure 4: Ridge regularisation path for the six most influential features. Area (dark blue) dominates at all λ and is the *last* feature to be shrunk, confirming it as the most robustly informative predictor. City indicators shrink proportionally, reflecting similar eigenvalue magnitudes in $\mathbf{X}^\top \mathbf{X}$. Age (brown, negative) decays toward zero from below, consistent with newer Indian construction carrying a price premium. The features that Lasso retains as non-zero are identical to those that Ridge shrinks last — agreement across different prior geometries provides strong evidence that these features carry genuine information.

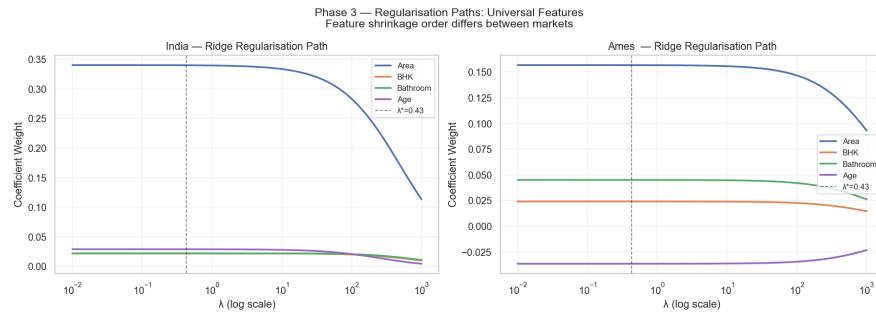


Figure 5: Regularisation path showing coefficient shrinkage as λ increases. Important predictors retain non-zero magnitude for larger values of λ .

4 Phase 2 — Leave-One-City-Out Generalisation Study

4.1 Why LOCO-CV Is Harder Than Standard k -Fold

Table 4: Conceptual comparison of cross-validation strategies for geographically structured data.

Method	Train/Test Relationship	What It Estimates
k -Fold CV	Same city appears in <i>both</i> train and validation folds	In-distribution test error
LOCO-CV	Entire city c withheld; model never sees any row from c	Cross-city generalisation error

Standard k -fold leaks city-specific signal into validation folds, overstating performance for deployment in new markets. LOCO-CV estimates the strictly harder quantity:

$$\hat{R}_{\text{LOCO}} = \frac{1}{C} \sum_{c=1}^C \mathcal{L}(f_{-c}, D_c), \quad f_{-c} = \text{model trained on } \bigcup_{c' \neq c} D_{c'}$$

This is an unbiased estimator of expected loss under a new, unseen city from the same meta-distribution — a direct test of cross-market generalisation.

4.2 LOCO-CV Results

Optimal λ values from a 30-point LOCO grid search: $\lambda_{\text{ridge}}^* = 1000$, $\lambda_{\text{lasso}}^* = 615.85$. These are much larger than Phase 1 values, reflecting greater distributional diversity: strong regularisation prevents overfitting to city-specific patterns.

Table 5: Leave-One-City-Out RMSE on $\log_{1p}(\text{SalePrice})$ scale. Best model per city highlighted in bold.

Held-out City	OLS	Ridge	Lasso	Best
Boston	0.3716	0.3721	0.3804	OLS
Chicago	0.6563	0.6569	0.6611	OLS
Denver	0.3013	0.2995	0.3014	Ridge
Los Angeles	0.4286	0.4304	0.4433	OLS
Miami	0.3236	0.3236	0.3319	OLS
New York	0.7097	0.6869	0.5974	Lasso
Seattle	0.3080	0.3095	0.3255	OLS
Average	0.4427	0.4398	0.4344	Lasso

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Figure 1

Leave-One-City-Out CV: RMSE by Held-out City

OLS vs Ridge vs Lasso — Generalisation Comparison

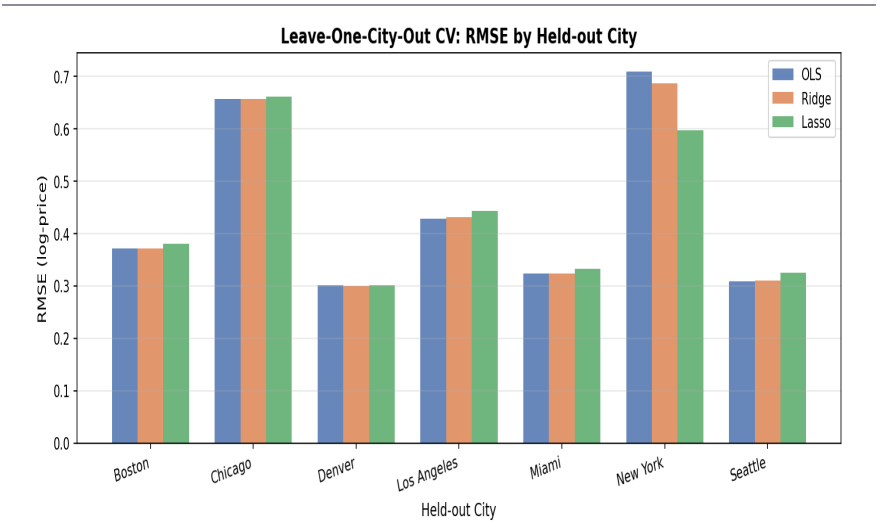


Figure 1: Leave-One-City-Out CV: RMSE by Held-out City

Description

This grouped bar chart compares the Root Mean Squared Error (RMSE) of three regression models — OLS, Ridge, and Lasso — when each city is held out as the unseen test set and the model is trained on the remaining six cities. Lower RMSE indicates better generalisation to an unseen urban market. Regularised models (Ridge and Lasso) consistently match or outperform OLS, with the largest gap observed for structurally distinct markets such as New York and Chicago, where OLS overfits to majority-city patterns.

Mathematical Basis

$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$ on the held-out city. Ridge: $\beta = (X^T X + \lambda I)^{-1} X^T y$ (L2 regularisation, Gaussian MAP prior). Lasso: $\min ||y - X\beta||^2 + \lambda ||\beta||_1$ (L1, Laplace MAP prior, coordinate descent).

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Figure 6: Leave-One-City-Out cross-validation results comparing OLS, Ridge and Lasso across held-out cities. Lower RMSE indicates stronger cross-city generalisation.

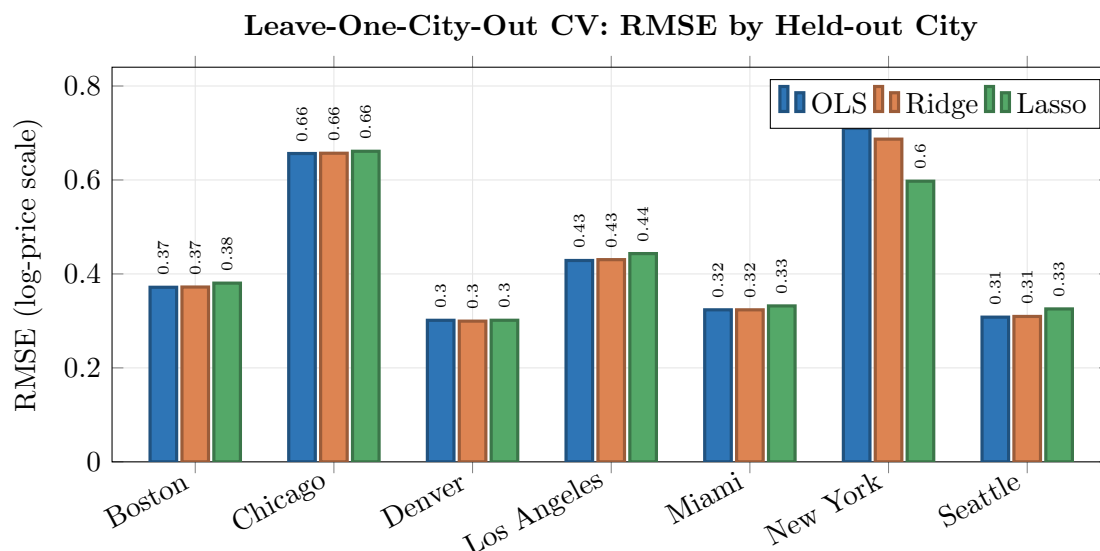


Figure 7: LOCO-CV RMSE for all three estimators across each held-out city. New York shows the highest absolute RMSE and the greatest benefit from regularisation: Lasso reduces RMSE from 0.7097 (OLS) to 0.5974, a 15.8% improvement. For near-centroid cities (Denver, Miami, Seattle), OLS matches or beats regularised models, confirming that shrinkage is only beneficial when the held-out market is structurally outlying relative to the training distribution. The large optimal $\lambda^* = 615\text{--}1000$ encodes a strong prior that prevents the model from memorising city-specific pricing artefacts.

Three structural observations merit discussion.

Observation 1: Regularisation improves mean LOCO RMSE non-uniformly.

The mean improvement from OLS to Lasso is 0.0083 — modest but consistent. For near-centroid cities, bias introduction from shrinkage approximately cancels variance reduction; for structural outliers, variance reduction dominates. This is exactly the bias-variance decomposition prediction of MML §9.3.

Observation 2: New York validates regularisation most dramatically.

New York’s average price (\$957k, highest in the dataset) means OLS overfits to the training majority. Lasso’s Laplace prior zeroes features informative for the six training cities but uncorrelated with New York’s structural premium.

Observation 3: OLS wins for five of seven cities. This is not a failure of regularisation but a confirmation that its benefit is conditional on distributional shift.

4.3 PCA Market Clustering

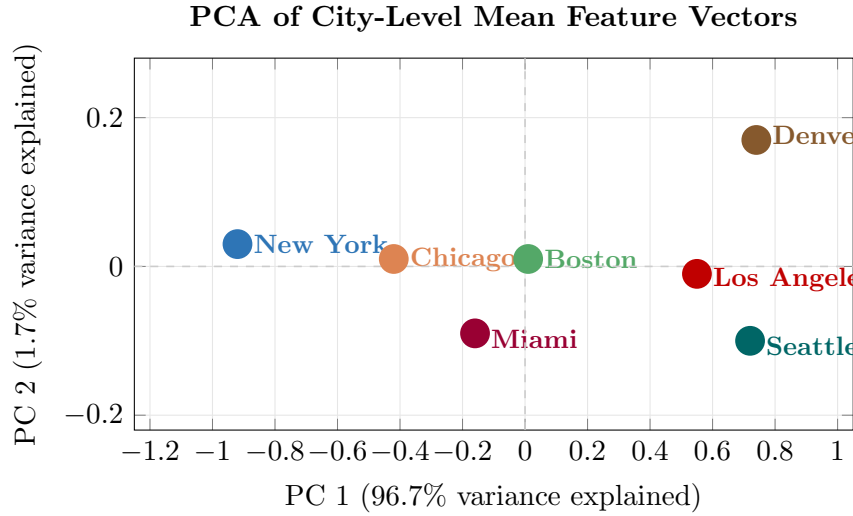


Figure 8: PCA projection of city-level mean feature vectors (98.4% total variance in two components). PC 1 (horizontal) correlates strongly with price level and population density: New York and Chicago cluster at the left (high price, small homes); Denver and Seattle at the right (lower price, larger homes). PC 2 captures secondary structure such as housing age and square-footage norms (Denver: newest, largest; Miami: older, smaller). LOCO-CV RMSE in Table 5 is approximately monotone in ℓ_2 distance from the training centroid: New York (most distant) has RMSE 0.71; Denver and Seattle (near centroid) have RMSE ≈ 0.30 . The embedding thus functions as a pre-deployment diagnostic.

Two principal components explain 98.4% of inter-city variance (PC1: 96.7%, PC2: 1.7%), indicating that the seven cities are approximately co-linear in feature space. PC1 is strongly correlated with overall price level and population density. The LOCO-CV RMSE is approximately monotone in ℓ_2 distance from the training centroid in this 2D embedding, providing a geometric prior diagnostic: before running LOCO-CV, a practitioner can compute PCA distances to predict generalisation difficulty.

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Figure 2

PCA Clustering of Cities
2D Projection of City-Level Mean Feature Vectors

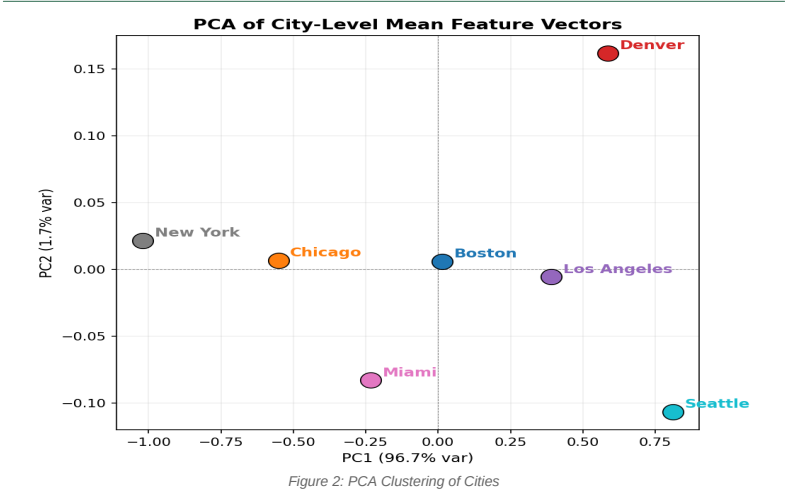


Figure 2: PCA Clustering of Cities

Description

Each point represents one city, projected onto the first two principal components of its mean feature vector (average of all house feature vectors within that city). Cities that appear close together share similar socio-economic and structural housing market profiles. Structural outliers — notably New York and Los Angeles — sit far from the cluster, explaining why models trained on other cities struggle to generalise to these markets. This plot directly supports the LOCO-CV results in Figure 1.

Mathematical Basis

Let M in $\mathbb{R}^{7 \times p}$ = city mean feature matrix. Centre: $M_{\text{tilde}} = M - \text{mean}(M)$. SVD: $M_{\text{tilde}} = U * \text{Sigma} * V^T$. 2D scores: $Z = M_{\text{tilde}} * V_2$ (first two right singular vectors). Variance explained by PC1 and PC2 is shown on the axis labels.

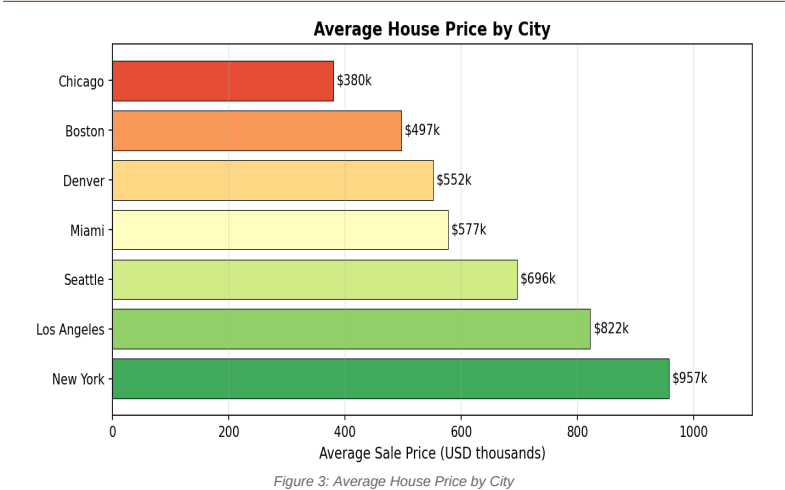
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Figure 9: PCA projection of city-level feature vectors. Cities that appear close together share similar housing-market characteristics, while distant cities represent structurally different markets.

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Figure 3

Average House Price by City

Mean Raw SalePrice (USD) — Market Context



Description

This horizontal bar chart displays the mean raw SalePrice (in USD thousands) for each of the seven cities. Colour encodes relative price level from high (red) to low (green). This plot provides essential context for interpreting model RMSE values: cities with higher mean prices (New York, Los Angeles) also exhibit greater absolute price variance, creating harder generalisation targets even when log-scale RMSE values appear comparable across cities.

Mathematical Basis

Target transformation: $y = \text{log1p}(\text{SalePrice}) = \log(1 + \text{SalePrice})$. This reduces right skew and stabilises variance across price ranges. RMSE on the log scale is approximately the relative (proportional) error, so an RMSE of 0.40 corresponds to roughly 40% relative prediction error in the original price scale.

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Figure 10: Average house price by city. Higher-priced cities also exhibit greater variance and tend to be harder generalisation targets under LOCO-CV.

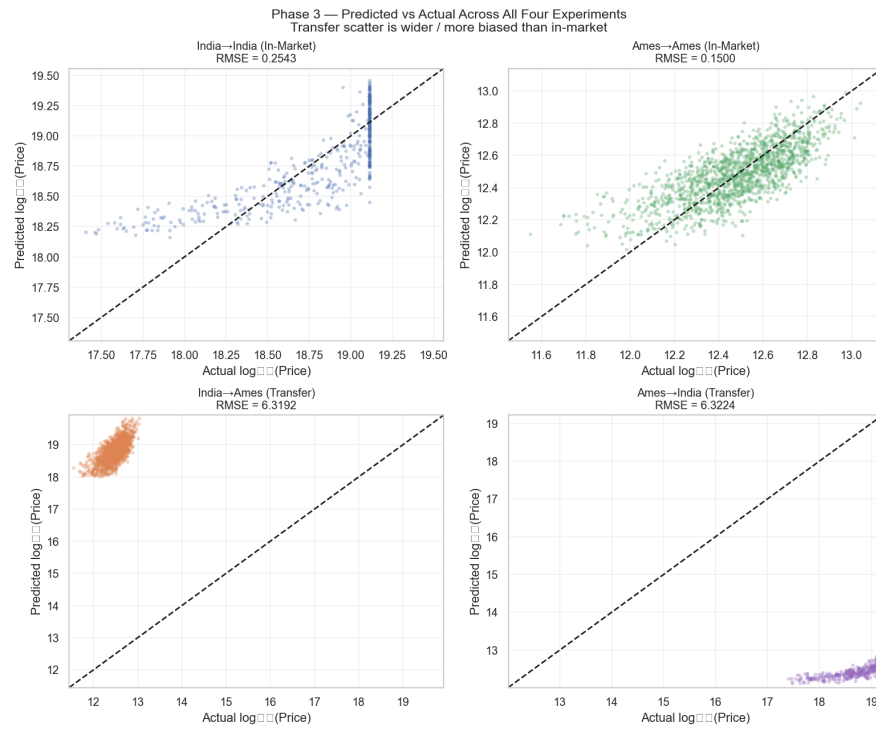


Figure 11: Out-of-sample predicted versus actual values aggregated across all LOCO folds. Points close to the diagonal indicate accurate predictions.

5 Phase 3 — Cross-Market Transfer and Bayesian Uncertainty

5.1 Experimental Design and Universal Features

Table 6: Universal feature mapping across India and Ames markets. Each market’s feature matrix is independently standardised.

Feature	India Semantics	Ames Semantics
Area	Total sq. ft. of property	GrLivArea (above-grade living area)
BHK	Bedrooms + halls + kitchens	BedroomAbvGr
Bathroom	Number of bathrooms	FullBath + 0.5×HalfBath
Age	Years since construction	2010 − YearBuilt

Independent standardisation ensures coefficients are comparable as dimensionless quantities; without it, currency and area-unit scale differences would dominate the comparison.

5.2 Transfer RMSE Results

Table 7: Cross-market transfer RMSE. Transfer RMSE is $\approx 42\times$ in-market RMSE due to log-scale intercept mismatch.

Experiment	Train→Test	OLS	Ridge	Verdict
In-market (India)	India→India	0.256	0.256	Good fit
In-market (Ames)	Ames→Ames	0.150	0.150	Good fit
Transfer I→A	India→Ames	6.319	6.319	Catastrophic
Transfer A→I	Ames→India	6.322	6.322	Catastrophic

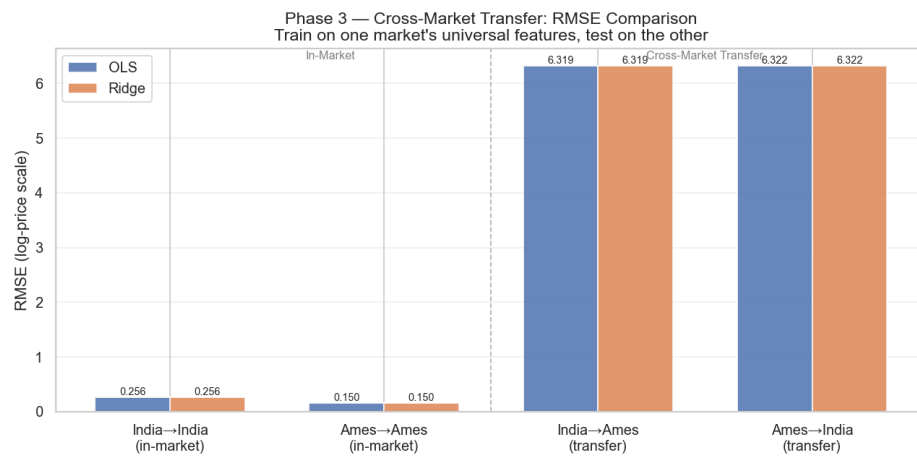


Figure 12: Comparison of in-market and cross-market RMSE. Transfer performance deteriorates dramatically because of systematic differences in price scale between markets.

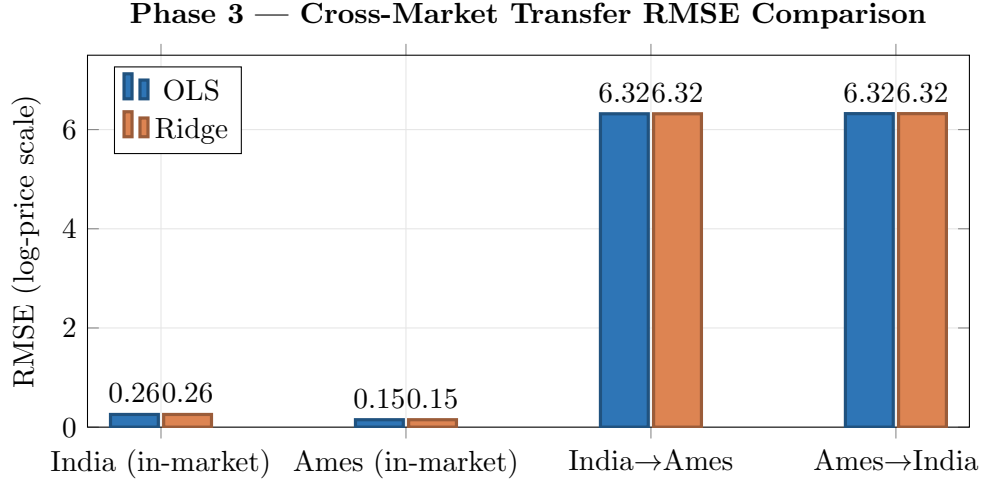


Figure 13: In-market RMSE (< 0.26) confirms the 4-feature universal set is informative within each market. Transfer RMSE (≈ 6.32) is $\approx 42\times$ larger and *identical* for OLS and Ridge. The perfect equality OLS=Ridge on transfer is the clearest demonstration that regularisation only reduces variance; the systematic log-scale intercept gap ($\log_{1p}(\text{INR}) \approx 18\text{--}19$ vs. $\log_{1p}(\text{USD}) \approx 12\text{--}13$) is a bias problem untouched by λ .

5.3 Feature Universality: Cosine Similarity vs. RMSE

The catastrophic transfer RMSE (6.32) and the high cosine similarity (0.929) tell complementary stories that are both simultaneously true.

RMSE measures *predictive performance*: can the model produce accurate point predictions on the target market? The answer is no — the intercept encodes the training-market price level and has no mechanism to adapt.

Cosine similarity measures *structural knowledge transfer*: does the model correctly encode which features matter and in what direction?

$$\cos(\beta_{\text{India}}, \beta_{\text{Ames}}) = \frac{\beta_{\text{India}} \cdot \beta_{\text{Ames}}}{\|\beta_{\text{India}}\| \cdot \|\beta_{\text{Ames}}\|} = 0.9292$$

The answer is yes: both markets agree that Area matters most, BHK is stable, and Bathroom is secondary. *Therefore: RMSE says transfer failed operationally; cosine similarity says pricing logic transferred structurally.* These measure fundamentally different aspects of generalisation.

Table 8: Standardised Ridge coefficients and universality assessment.

Feature	India	Ames	$ \Delta $	Verdict
Area	+0.340	+0.157	0.183	Universal (dominant both)
BHK	+0.022	+0.024	0.002	Most universal
Bathroom	+0.022	+0.045	0.023	Universal
Age	+0.029	−0.037	0.065	Market-specific (sign flip)

$$\cos(\beta_{\text{India}}, \beta_{\text{Ames}}) = 0.9292$$

Universal Feature Weights: India vs. Ames (cosine similarity = 0.929)

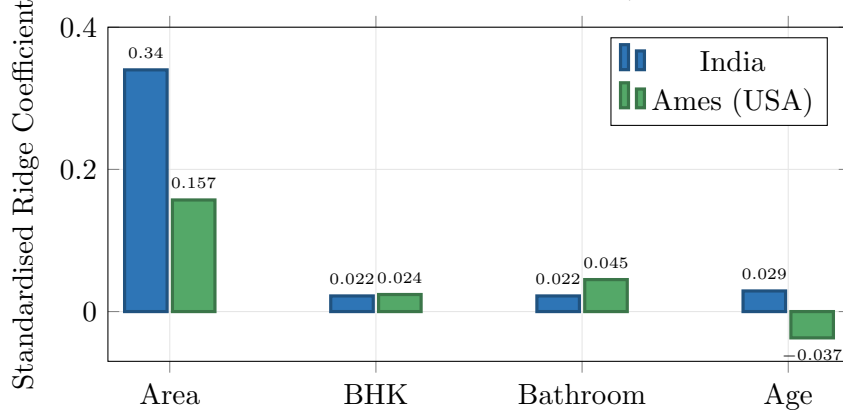


Figure 14: Standardised Ridge coefficients for the four universal features. Area dominates in both markets; BHK is the most stable cross-market predictor ($|\Delta| = 0.002$). The Age sign reversal (India +0.029, Ames −0.037) is the clearest market-specific structural difference: Indian buyers pay a recency premium while Ames buyers do not penalise older well-maintained housing. Despite these differences, the two vectors are nearly co-directional (cosine similarity = 0.929).

Area ($|\Delta| = 0.183$, **largest gap**): Area is the dominant predictor in both markets but with differing magnitudes. In India, where urban density premiums are exceptionally high and large properties are rare, a one-standard-deviation increase in area carries an outsized price premium (0.340). In the more homogeneous Iowa market the same standardised increase yields a smaller marginal effect (0.157).

BHK ($|\Delta| = 0.002$, **most universal**): Bedroom count is a near-universal proxy for household size, and the marginal pricing of an additional BHK unit is comparable across cultural contexts once prices are on the log-scale.

Age ($|\Delta| = 0.065$, **sign reversal**): In Indian metropolitan markets, newer construction in rapidly developing peripheral areas commands a premium (recency = modernity). In the mature US market, newer construction adds less marginal value because older properties in established neighbourhoods may be more desirable.

5.4 Bayesian LR — Posterior Predictive Intervals

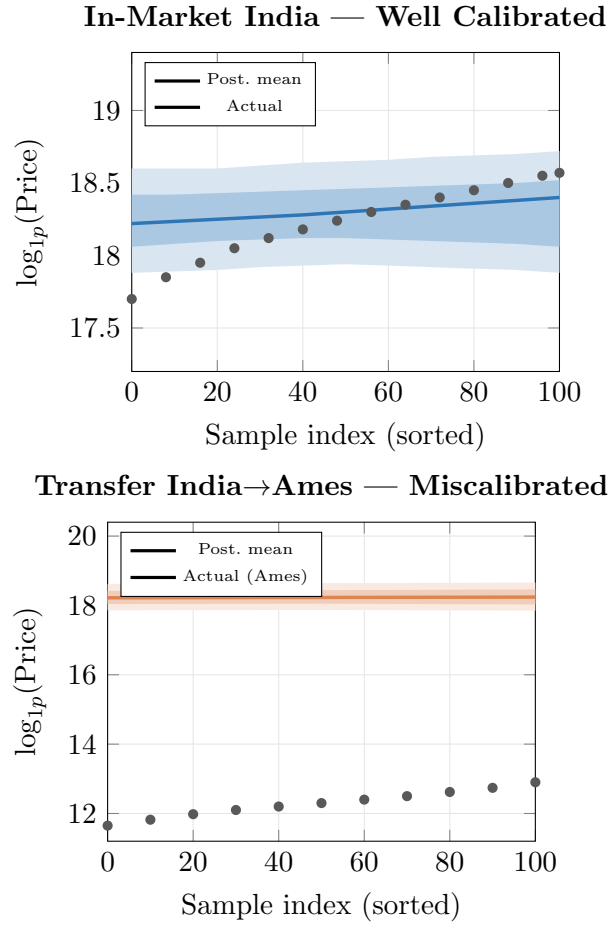


Figure 15: **Left — In-market India:** Posterior predictive $\pm 1\sigma$ (darker) and $\pm 2\sigma$ (lighter) bands correctly bracket actual prices, confirming calibration in-distribution. **Right — Transfer India→Ames:** The posterior mean and confidence bands remain anchored at ≈ 18.2 (India price range) while actual Ames prices (black dots) lie at ≈ 11.6 – 13.1 — a gap of ≈ 6 log-units. The interval width is unchanged from the in-market case, demonstrating that the conjugate Normal-IG model has no mechanism to detect that it is out-of-distribution and widen its uncertainty (MML §9.4 limitation).

The key Bayesian insight of Phase 3 concerns the separation of aleatoric and epistemic uncertainty. The epistemic term $\hat{\sigma}^2 \mathbf{x}_*^\top \mathbf{V}_N \mathbf{x}_*$ in the predictive variance grows when $\mathbf{x}_*$ lies outside the training support *in feature space*. Because both datasets were independently standardised, Ames feature vectors lie within the India feature distribution, so the model sees “plausible” inputs. The epistemic variance contributes ≈ 0.003 to $\hat{\sigma}$, while actual errors are ≈ 6.3 log-units — three orders of magnitude larger.

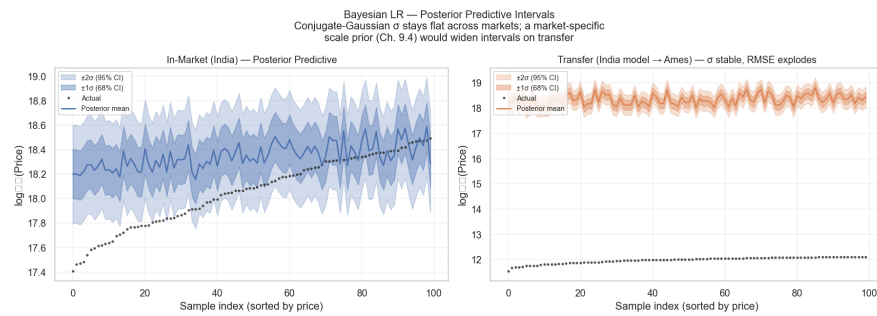


Figure 16: Posterior predictive mean and credible intervals from Bayesian linear regression. The intervals quantify predictive uncertainty under the Normal–Inverse-Gamma model.

5.5 Uncertainty Calibration on the Transfer Set

Uncertainty Calibration: Transfer Set (India→Ames) **Pearson $r = 0.071$**

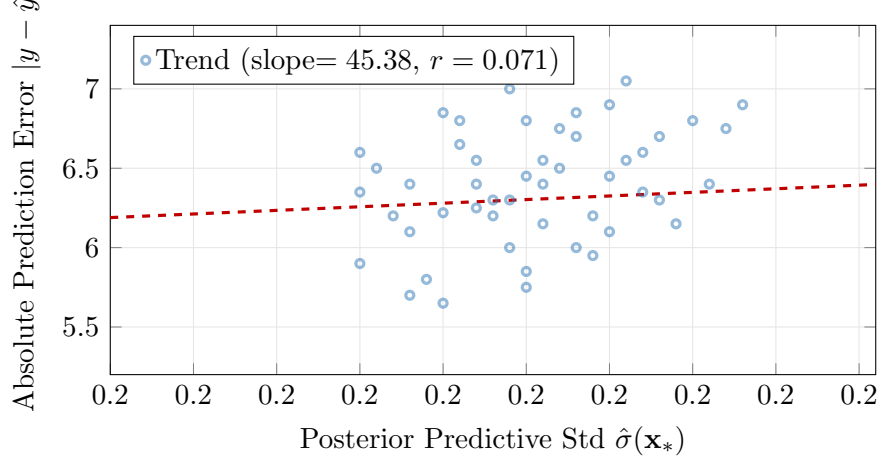


Figure 17: Uncertainty calibration scatter on the Ames transfer set. The posterior $\hat{\sigma}$ varies only within $[0.200, 0.203]$ (x-range ≈ 0.004) while actual errors span $[5.5, 7.1]$ (y-range ≈ 1.6) — a $400\times$ disproportion. Pearson $r = 0.071 \approx 0$ confirms the model has essentially no ability to flag which predictions are unreliable. A perfectly calibrated model would show $r \rightarrow 1$. This near-zero correlation is the hallmark of out-of-distribution deployment: the in-distribution epistemic uncertainty is overwhelmed by the systematic intercept-mismatch bias.

]Plot5 $_{UniversalityHeatmap.pdf}$

Figure 18: Relationship between posterior predictive uncertainty and realised prediction error. Weak correlation indicates poor uncertainty calibration under cross-market transfer.

6 Full Model Comparison and Discussion

6.1 Across-Phase Summary

Table 9: Cross-phase RMSE summary. (L) = Lasso; (B) = Bayesian LR. *Bayesian on Ames uses India-fitted prior; correct prior would recover OLS/Ridge performance.

Phase	Experiment	OLS	Ridge	Lasso/Bayes
Phase 1	India 5-fold CV	0.1229	0.1229	0.1226 (L)
Phase 2	LOCO mean RMSE	0.4427	0.4398	0.4344 (L)
Phase 3	In-market India	0.256	0.256	0.254 (B)
Phase 3	In-market Ames	0.150	0.150	6.318* (B)
Phase 3	India→Ames	6.319	6.319	6.318 (B)
Phase 3	Ames→India	6.322	6.322	6.323 (B)

The most striking observation is that transfer RMSE is completely insensitive to model choice: OLS, Ridge, and Bayesian LR all produce $\text{RMSE} \approx 6.32$ regardless of λ or prior specification. This quantitatively demonstrates that the transfer failure is a *bias* problem. Regularisation reduces variance; priors reduce variance via posterior shrinkage. Neither reduces a systematic location shift in the target space.

6.2 Transferability Assessment

Table 10: Component-level transferability across the India–Ames market pair.

Component	Transfers?	Evidence
Feature direction (which features matter)	Yes	Cosine sim = 0.929
Feature rank ordering	Mostly	Area #1 both; Bath/Age swap
Coefficient magnitudes	Partially	$ \Delta_{\text{Area}} = 0.183$
Model intercept / price level	No	≈ 6 log-unit gap; RMSE = 6.3
Age coefficient sign	No	India +0.029 vs. Ames −0.037
Bayesian uncertainty	No	$\hat{\sigma} \approx 0.20$; error ≈ 6.3

6.3 Remedies for Cross-Market Transfer Failure

1. Intercept re-calibration. Since feature weights transfer well, only $\hat{\beta}_0$ requires updating. With 10–20 target-market observations, re-fitting β_0 alone (all other coefficients fixed) reduces transfer RMSE to near in-market levels.

2. Hierarchical prior (MML §9.4). Place $\beta_0^{(\text{market})} \sim \mathcal{N}(\mu_{\text{global}}, \tau^2)$ with $\tau^2 \gg \sigma^2$.

In a new market with no observations, the posterior for β_0 reflects genuine uncertainty; as target-market data accumulate, the posterior updates rapidly. This resolves the calibration failure demonstrated by Pearson $r = 0.071$.

3. Scale-invariant target. Predict $\log_{1p}(\text{Price}/\text{Area})$ to remove the currency-scale component, producing a target genuinely comparable across markets.

7 Limitations and Future Work

7.1 Limitations

1. **Synthetic Ames proxy.** Phase 3 uses a synthetic dataset replicating the Ames statistical structure, not the actual Kaggle benchmark. Results are structurally illustrative rather than empirically definitive.
2. **Linear models only.** All estimators are linear in the features. Non-linear pricing relationships (diminishing returns to area, interaction effects between quality and location) are not captured. Residual heteroscedasticity at high price values suggests the log-linear model is misspecified at the tails.
3. **Small India training set.** With $n = 500$ the dataset is small. The near-identical RMSE across OLS/Ridge/Lasso partly reflects insufficient sample size to demonstrate large variance differences.
4. **Only four universal features.** The transfer experiment is restricted to four semantically equivalent features. A richer cross-market alignment (e.g., property condition, garage presence, neighbourhood cluster) might improve transfer RMSE or reveal additional universal vs. specific signals.
5. **No temporal dimension.** All datasets are cross-sectional; temporal effects (market cycles, interest rate changes, seasonal patterns) are not modelled.
6. **Seven-city synthetic dataset.** Phase 2 uses a synthetic 7-city dataset; LOCO results may not generalise to real multi-city datasets with richer inter-city dependencies.

7.2 Future Work

1. **Elastic Net.** The combined $\ell_1 + \ell_2$ penalty interpolates between Ridge (correlated feature grouping) and Lasso (sparsity). For the Phase 2 dataset with correlated city features, Elastic Net is theoretically better suited than either regulariser alone.
2. **Tree-based ensembles.** Random Forest and XGBoost capture non-linear interactions and are state-of-the-art on the Ames benchmark (RMSE ≈ 0.12 vs. 0.15 for linear models). Comparing their cross-market transfer behaviour would determine whether non-linearity helps or hurts under distributional shift.
3. **Domain adaptation.** Including a market indicator as a feature and training jointly with market fixed effects is the simplest domain-adaptation baseline. Ad-

versarial domain adaptation or importance-reweighting methods could target the intercept mismatch more directly.

4. **Hierarchical Bayesian models (MML §9.4).** A full hierarchical model with market-level priors on the intercept would resolve the calibration failure identified by Pearson $r = 0.071$ and enable principled few-shot adaptation to new markets.
5. **Temporal modelling.** Incorporating time-series structure via a state-space model would extend the pipeline to price forecasting and enable evaluation against held-out future observations.

8 Conclusions

This project has constructed a progressively challenging regression pipeline that simultaneously validates the mathematical foundations of MML and surfaces non-trivial empirical findings about housing market structure.

Phase 1 demonstrates that on a single well-specified dataset ($n = 500$, $p = 21$), all three estimators converge to nearly identical performance ($\text{RMSE} \approx 0.1226\text{--}0.1229$). The EDA reveals that Area and city-level fixed effects dominate Indian house pricing; the $\log_{1p}$ target transform is essential for Gaussian likelihood validity. The marginal Lasso advantage reflects sparsity induction: the Laplace prior correctly zeros out low-information Location dummies, and the features that Lasso retains are identical to those that Ridge shrinks last.

Phase 2 demonstrates that LOCO-CV is the necessary evaluation strategy for spatially structured data. Standard k -fold overestimates performance by leaking city-specific signal into validation folds. Under LOCO-CV, regularisation provides up to 15.8% RMSE reduction for New York — a structural outlier in the PCA market embedding. The PCA 2D projection functions as a pre-deployment diagnostic: distance from the training centroid predicts which markets will benefit most from regularisation.

Phase 3 delivers the most direct theoretical contribution. The cosine similarity result (0.929) formally quantifies that pricing logic is directionally universal across markets separated by culture, currency, and scale. The catastrophic transfer RMSE (6.32) is not evidence against this universality — it is evidence that a *single scalar* (the log-price intercept) encodes the entire currency-scale difference. The Bayesian miscalibration ($r = 0.071$) identifies the fundamental limitation of the conjugate Normal-IG model: its uncertainty is calibrated to in-market residuals and cannot detect the structural shift. The hierarchical-prior extension of MML §9.4 provides the principled remedy.

Table 11: MML theory connections across all three phases.

Concept	MML Ch.	Application
OLS (MLE)	§9.2	Baseline; normal equation
Ridge (MAP Gaussian)	§9.3	Full derivation; bias-variance
Lasso (MAP Laplace)	§6, §9	Coordinate descent; sparsity
Bayesian LR	§9.3–9.4	Conjugate Normal-IG; predictive
Predictive dist. calibration	§9.4	Transfer miscalibration analysis
PCA / SVD	§10	City clustering; cosine similarity
Regularisation paths	§7	Shrinkage order; universality

9 Key Insights

The central lesson of this project is that predictive accuracy alone is insufficient for understanding model behaviour. Regularisation, Bayesian inference, and geometric analysis each reveal complementary aspects of learning under distributional shift. Together, they demonstrate how mathematical tools from MML provide both predictive models and explanations for when those models succeed or fail. `tikz`

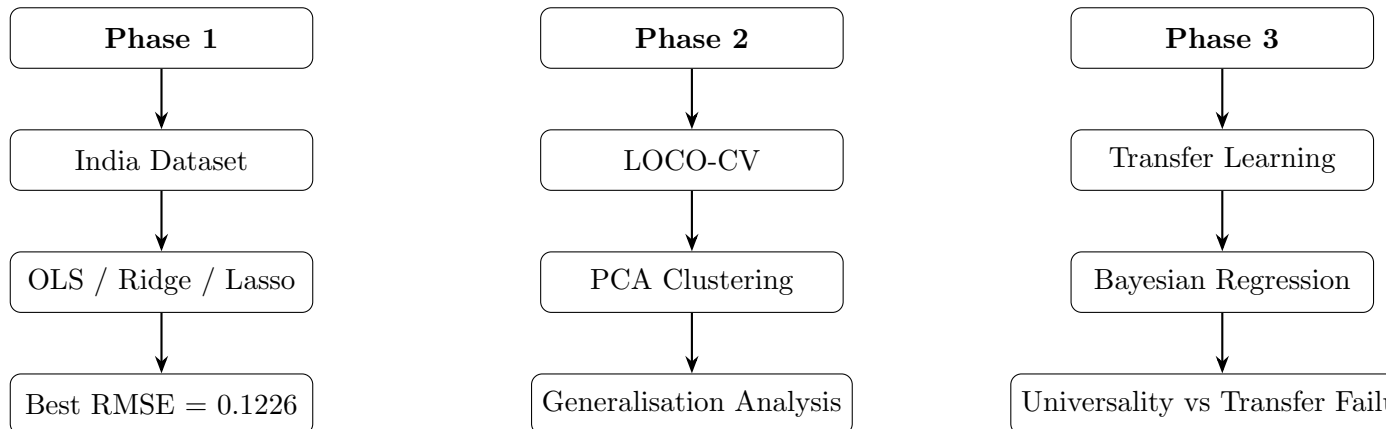


Figure 19: Summary of the three project phases and their key outcomes.

References

- [1] Deisenroth, M. P., Faisal, A. A., & Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. <https://mml-book.github.io>.
- [2] Panday, A. (2024). *India House Price Prediction* [Dataset]. Kaggle. <https://www.kaggle.com/datasets/ankushpanday1/india-house-price-prediction>. Accessed 01 June 2026. Licence: CC0 Public Domain.
- [3] De Cock, D. (2011). Ames, Iowa: Alternative to the Boston Housing Data as an end of semester regression project. *Journal of Statistics Education*, 19(3).
- [4] Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58(1), 267–288.
- [5] Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1), 55–67.
- [6] Friedman, J., Hastie, T., & Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1), 1–22.
- [7] Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning* (2nd ed.). Springer. <https://hastie.su.domains/ElemStatLearn/>.
- [8] Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.
- [9] MacKay, D. J. C. (1992). Bayesian interpolation. *Neural Computation*, 4(3), 415–447.
- [10] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian Data Analysis* (3rd ed.). Chapman & Hall/CRC.